Review 6 (HW6)
Forces and Torques

10/19/2026

- **Energy:**

$$W_m(\lambda, x) = \int_0^{\lambda} i(\hat{\lambda}, x|_{const})\, d\hat{\lambda}$$

$$i = \frac{\partial W_m(\lambda, x)}{\partial \lambda}\bigg|_{x=const}$$

$$f^e = -\frac{\partial W_m(\lambda, x)}{\partial x}\bigg|_{\lambda=const}$$

- **Co-energy:**

$$W_m'(i, x) = \int_0^{i} \lambda(\hat{i}, x|_{const})\, d\hat{i}$$

$$\lambda = \frac{\partial W_m'(i, x)}{\partial i}\bigg|_{x=const}$$

$$f^e = +\frac{\partial W_m'(i, x)}{\partial x}\bigg|_{i=const}$$

* $x \to \theta$, $f^e \to T^e$

* $W_m + W_m' = i\lambda$

If electrically linear system:

$$\lambda(i, x) = L(x) \cdot i$$

$$W_m(\lambda, x) = \frac{\lambda^2}{2L(x)}$$

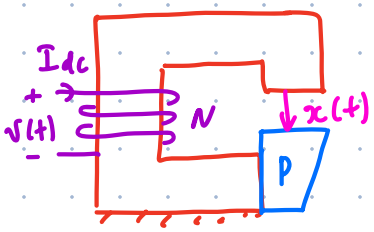
$$W_m'(i, x) = \frac{L(x) \cdot i^2}{2}$$

$$W_m = W_m' \text{ (numerically)} = \frac{1}{2}\lambda i = \frac{L(x) i^2}{2} = \frac{\lambda^2}{2L(x)}$$

① Fall 2019, M2, pr 1

A magnetic structure is shown below, with a movable member P. The area of the air gap is A, and the size of the gap is x(t). The magnetic structure is excited by a coil of N turns carrying a positive, constant current i(t) = Idc > 0, in the polarity shown.

Neglect fringing and assume that the permeability of the structure to be infinite. Recall that the permeability of free-space is $\mu_0 = 4\pi \times 10^{-7}$ [H/m] by definition.



upward / downward
trying to close the gap / opposite to the direction x / want to minimize the reluctance

a) What are the magnitude and direction of the force of electric origin acting on the member P? State your answer as a function of μ_0, A, x(t), N, and Idc. State the unit for force. (9 points for magnitude, 0.5 point for correct direction, 2.5 points for justification of correct direction.)

Via co-energy:

① $f^e = \frac{\partial W_m'(i,x)}{\partial x}$

② $W_m'(i,x) = \int_0^i \lambda(\vec{i},x)\, d\vec{i}$ or if linear $W_m'(i,x) = \frac{L(x)\cdot i^2}{2}$?

$\Downarrow$

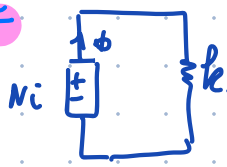
$W_m'(i,x) = \frac{N^2\mu_0 A \cdot i^2}{2x}$

$\Downarrow$

$f^e = -\frac{N^2\mu_0 A}{2x^2}\cdot I_{dc}^2$ [N]

" $|f^e|$

MEC:



$\lambda = N\phi = \frac{N^2\mu_0 A}{x} i$ ($L(x)$)

el. linear

Approach 2 (Not recommended): via W_m:

① $W_m(\lambda,x) = \frac{\lambda^2}{2L(x)} = \frac{x\lambda^2}{2\mu_0 A N^2}$

② $f^e = -\frac{\partial W_m(\lambda,x)}{\partial x} = -\frac{\left(\frac{N\mu_0 A}{x}\cdot I_{dc}\right)^2}{2\mu_0 A N^2} = -\frac{\mu_0 A N^2}{2x^2} I_{dc}^2$ [N]

substitute here!

② Spring 2015, M2, pr 3

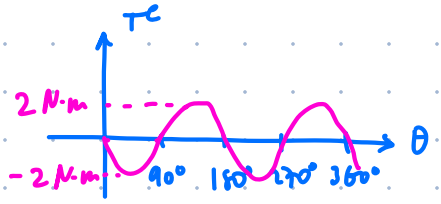
Consider a reluctance rotating motor of a single coil, with the inductance sinusoidal on the rotor position:

$L(\theta) = 1 + 2\cos(2\theta)$

The flux linkage is linear in current i. $\leftarrow \lambda = L(\theta)\cdot i \Rightarrow W_m'(i,\theta) = \frac{L(\theta)\cdot i^2}{2} = \frac{(1+2\cos 2\theta)\, i^2}{2}$

a) Find the maximum possible torque when $i = 1$ Ampere;

① $T^e(i,\theta) = \left.\frac{\partial W_m'(i,\theta)}{\partial \theta}\right|_{i=\text{const}} = -2\sin(2\theta)\cdot i^2 = -2\sin(2\theta)$ ($i^2 \uparrow 1A$)



$|T^e_{max}| = 2$ Nm

③ Fall 2015, M2, pr 1

The co-energy of a device is given by

$$Wm'(i,x) = \frac{i^3}{6x} + \frac{i}{x}$$

Find:

a) The expression for flux linkage (λ) as a function of i and x. ✓ $\leftarrow (\lambda) = \left.\frac{\partial W_m'(i,x)}{\partial i}\right|_{x=\text{const}} = \frac{3i^2}{6x} + \frac{1}{x} = \left(\frac{i^2}{2x} + \frac{1}{x}\right)$

b) The force of electrical origin $f^e(i,x)$ $\leftarrow f^e = \left.\frac{\partial W_m'(i,x)}{\partial x}\right|_{i=\text{const}} = -\frac{i^3}{6x^2} - \frac{i}{x^2} = -\frac{1}{x^2}\left(\frac{i^3}{6} + i\right)$

c) The energy stored in the coupling field Wm as a function of i and x.

$W_m + W_m' = i\lambda \Rightarrow W_m(i,x) = i\lambda - W_m'(i,x) = i\left(\frac{i^2}{2x} + \frac{1}{x}\right) - \frac{i^3}{6x} - \frac{i}{x} = \frac{i^3}{3x}$

non-linear! ⇓ $W_m \neq W_m'$